Task 5: ***Answer the following questions briefly and clearly***

How does this implementation differ from the traditional DSW approach?

Traditionally, the DSW algorithm converts the tree into a right leaning vine, essentially making it a single linked list. Phase 1's change to a left leaning vine is really just a mirror of this but also seeks to make it a bit more efficient by only rotating if the subtree size is <= 2 nodes.

What is the impact of phase 1

Phase 1 converts the tree into a mostly left skewed vine, with occasionally right children with small subtree size that did not get rotated because of the above.

What is the impact of phase 2

Phase 2 performs right rotations on the left skewed vine in order to balance the tree into a structure with height close to 2log2(N) where N is the number of nodes.

What tradeoff did you observe

The tradeoff of only performing rotations on small subtrees makes the structure slightly more efficient and less expensive, as less rotations are done to convert the tree into a left skewed vine. On the other hand, the position of the 'extra' nodes that still exist as right children in the vine state aren't uniform or defined as they are in the usual implementation of the DSW algorithm (being that the 'extra' nodes will be left aligned in the incomplete bottom layer of the tree).

Can you think of other variations. Explain this in detail and justify your design choices.

I variation I thought about during my work was that after creating a left/right skewed vine, we can perform a number of left/right rotations on the root such that the node in the middle of the vine is splayed to the root position. This would create a structure resembling a pyramid of sorts (artist depiction below). Following this, we know that the number of nodes in the left and right subtrees are more or less equal, and we can recursively continue breaking each subtree into the same shape until the structure is balanced.

To elaborate more using the visual aid below, the height of the vine was 7, so with 3 rotations the 'middle' node is splayed to the root. Then, for the left and right subtree, perform 3/2 = 1 rotations on them to fully balance the tree.